1 Types of information systems projects

Learning outcomes

When you have finished reading this chapter, you will be able to:

- List nine distinct types of IS project that may be encountered
- Describe two key characteristics of each of these nine types of IS project
- Describe the particular skills required by project managers to manage each of the types of IS project
- Describe how to tailor the project management approach for smaller IS projects.

1.1 Introduction

This book is called *Project Management for Information Systems* but, in fact, within the information systems field, there is a very wide variety of types of projects that may be undertaken. Although the general principles of project management are broadly common to all IS projects – indeed, to projects in all disciplines – there are nonetheless ways in which the different types of IS project do diverge from one another and which we need to address before getting into the detail of IS project management. We have grouped IS projects into nine broad types:

- Software development
- Package implementation
- System enhancement
- Consultancy and business analysis assignments
- Systems migration
- Infrastructure implementation
- Outsourcing (and in-sourcing)
- Disaster recovery
- Smaller IS projects.

Of course, there are other classifications we could have used – for example, into 'internal' projects (done within an organization) or 'external' (carried out under contract). It is also true that some IS projects embrace more than one of our nine categories; one involving package implementation, original software development and putting in a new infrastructure, for example. However, we believe that these nine basic categories cover most of the types of project likely to be encountered and enable us to explore the main differences and similarities between them.

In discussing the various types of project, we shall use the term 'supplier' to mean whoever is doing the work and 'customer' for whoever has commissioned or will benefit from that work. In the situation where, say, a consultancy firm is working under contract for a client, these terms are obviously applicable, but this is also the case where an in-house IT unit is doing the work for a departmental head.

In the rest of this book, we shall use a conventional software development project to illustrate how the project management principles and techniques are applied, but readers are advised to consider, at each stage, how the approach might need to be adapted if they are involved in one of the other main types.

1.2 Software development projects

This is what one usually thinks of first when considering IS projects. Essentially, we have a group of people working together to specify, design, develop, test and implement a new system for a 'customer' (either internal or external).

In principle, development projects have a lot in common with other types of construction work, and most of the traditional project management methods and techniques are applicable. However, one particular difficulty that does not generally face, say, construction project managers concerns the essential intangibility of a software system. With a bridge or a building, the design can be represented by blueprints and drawings and these are understandable by whoever has commissioned the project. It is usually quite hard for the users of a piece of software to express their wishes clearly and for the business analysts to capture them unambiguously. However hard all parties work at this, there are bound to be areas where supplier and customer have different ideas about what is to be done and where the specification proves to be ambiguous.

What this means is that the managers of software projects need:

- Flexibility of approach in being prepared to revisit the specification and negotiate with the customer as the project proceeds.
- Well-developed interpersonal and stakeholder-management skills.

The emergence of 'agile' development methods (see Chapter 6) is, in part, a response to the difficulty of specifying software requirements 'up front' and a recognition that a more evolutionary approach – where the suppliers and customers work closely together to refine their understanding of the need – may be a better way to deal with the inherent uncertainties and intangibilities of software development.

1.3 Package implementation projects

Buying a pre-existing software package and installing it represents an alternative, and usually quicker and cheaper, way of meeting customers' system requirements. In principle, package implementation is simple: the package is bought, installed, switched on and used – rather like buying and installing a television set or a microwave oven. However, there are some problems with acquiring a software package that do not generally apply to televisions and microwaves, namely:

- For the customer, the difficulty of selecting the correct package in the first place. The range of features in a TV or microwave are relatively few compared with a business system which is usually very complicated and has to cope with a myriad of 'exception situations' that are different from the way processes are supposed to operate. Thorough analysis of requirements is part of the answer but it has to be faced that there will remain some questions of detail that only emerge when the package is being tested or even after it has been installed and used.
- For the supplier, the problem of being asked to 'tweak' or adjust the package to match the customer's ways of working. Most packages provide for a degree of customization but some customers may have very specific ways of working that the package cannot easily be tailored to support. If the customer's expectations are not managed carefully, then the users of the package may be very disappointed when they get to use their new system.
- For both parties, the issue of integrating a package with other existing systems. It is very rare nowadays for any system to be completely free-standing, and integrating a package into an existing IT infrastructure can prove very complex. Very careful analysis of the integration requirements and detailed planning of the integration work are clearly vitally important here.

For the manager of a package implementation project, the main challenges are therefore:

- Managing the series of sub-projects – package customization and tailoring, data migration and cleansing, user training, cutover from old to new system – that is inevitably involved.
- Ensuring that the suppliers live up to any claims they have made in the sales process concerning the capabilities of their product and its suitability for the organization in which it is to be deployed.
- But also keeping the purchasers and users of the package realistic in their demands for changes and tailoring; are these really needed to fulfil a business need or do they result from a reluctance to change working practices to fit in with what the package can do?

The 'bottom line' is that buying a package is always likely to be a trade-off between the idealized demands of the system's end-users and what the organization can afford in terms of time, effort and money. Thus, apart from good planning skills, the project manager will require highly developed interpersonal skills, particularly in terms of supplier and customer management, negotiation and conflict resolution.

1.4 System enhancement projects

This type of project arises when the users, or owners, of an existing system want it enhanced to provide new features or functions or perhaps to meet some external demand, like compliance with legislation or regulations. Many such 'projects' are not recognized or managed as such, but are just handled as 'business as usual' by a support and maintenance team. However, large-scale enhancements need to be managed as real projects – for which see section 1.2 above – because, often, the amount of work, and thus of time and cost, is considerable and so should be subject to proper project-management discipline.

There are some particular issues that face the manager of a large enhancement project, including:

- The difficulty of keeping the existing system operational while work proceeds on the enhancement.
- The fact that the developers involved in the enhancement are often also engaged in supporting the system, when it can be hard to balance and reconcile the competing demands on their time.
- The need for rigorous 'regression testing' to ensure that the new enhancements do not damage parts of the existing system that were working well.

As ever, careful analysis of the requirements and thorough planning of the project – and particularly of the implementation aspects – are key to success in this type of project.

1.5 Consultancy and business analysis assignments

Some IS projects do not involve developing or installing anything tangible at all. Sometimes, they are about investigating a business issue and proposing solutions using information technology. Such consultancy and business analysis assignments are nevertheless projects, although they do pose some peculiar difficulties from a management perspective:

- We have already seen that software is, by its very nature, rather intangible and therefore difficult to estimate, plan for and control. This is even more so with consultancy and analysis work where the customer's need may be for someone to look into an issue where they (the customer) are not quite sure what the problem is or where suitable solutions may be found.
- Because of this, the budget and timescale for the project ought to be fairly flexible. However, often, the customer wants an 'answer' by some deadline and the problem now becomes one of expectation management; making

sure the customer understands the limitations of what can be achieved within this constraint.

- It is hard to fix the scope of consultancy projects – what is included and excluded. Indeed, the investigation work itself may reveal that the original boundaries of the project have been drawn too narrowly and will have to be expanded if worthwhile results are to be achieved. The scope will therefore have to be managed flexibly and carefully and the project manager needs to tread a fine line between, at one extreme, being seen to stick too rigidly to the original brief and, at the other, being accused of trying to expand the job, presumably to increase the value of the consultancy work.

1.6 Systems migration projects

This type of project is one where an existing operational system has to be moved to a new operating environment – perhaps because the current one is now longer supported or supportable. There may be some software development involved, because the new platform does not work exactly like the old one, and it may be necessary to create interfaces with other systems. There may also be infrastructure implications, for which see the next section, to consider. It might also be necessary to carry out some limited retraining of users to enable them to utilize the new environment.

From the point of view of the system's users, the project's success will be judged by the smoothness of the transition and the lack of interruption to their workload.

1.7 Infrastructure projects

This type of IS project includes ones to introduce or replace hardware, servers or PCs, for example, to put in place communications infrastructures and also sometimes the physical construction of things like computer suites or the fitting out and equipment of a new office building.

General project management principles are all applicable to this type of project and it does have the advantage, usually, that the outcomes of the project are nicely tangible – unlike, as we have seen, some other IS projects.

However, there are some issues that managers of this type of project need to consider, including:

- Usually, the need to maintain 'business as usual' whilst putting in place the new infrastructure. This can be tricky where, for example, there is limited space for old and new to sit alongside each other.
- Supplier management features heavily in these projects, as most of the work is usually subcontracted and all of the equipment is bought-in. It therefore becomes especially important to establish firm and realistic timescales for

delivery and to examine carefully the interdependencies between tasks, as otherwise time, effort and money can be wasted waiting around for things to be delivered or completed.

1.8 Outsourcing (and in-sourcing) projects

The reasons for outsourcing IT provision in an organization are many and various and have been discussed and debated widely over the past decade or so. They include:

- The wish to gain access to the pooled expertise of the outsourcing providers.
- Difficulty in managing the IT estate internally.
- A desire to reduce costs, through economies of scale or by taking advantage of lower labour costs elsewhere.
- The need to reduce employee head-count.
- A wish to put the provision of IT on the same basis as that of other essential ‘utilities’ such as gas or electricity.
- The belief that IT has become commoditized and no longer provides a source of competitive advantage.
- Long-term dismay by general managers over the costs of IT and the seeming impossibility of controlling it.

As we say, all of these are debatable and, in recent years, some organizations have gone in the opposite direction, taking back in-house IT provision that had formerly been outsourced. One case in point is a major UK supermarket chain which had outsourced its IT to one of the leading consultancy firms but became dissatisfied with the results and with the loss of control of what it came to see again as a key source of competitive advantage.

There is, sadly, an element of fashion in IT as in other fields and, sometimes, the in-sourcing/outsourcing decision may be affected as much by what other organizations are doing as by a provable business need.

Sometimes, it is not just the IT systems themselves that are outsourced but whole areas of business processing, including the systems that support these processes.

However, whatever the merits or otherwise of the situation, the fact remains that an outsourcing or in-sourcing *project* is one of the most complex that can be undertaken. This is because it will involve most or all of the following:

- The ‘due diligence’ work involved in making sure the scope of the contract is clear and feasible.
- Training new people in the IT systems to be supported.
- Also providing training in the business environment surrounding those systems.
- Taking inventories of assets – hardware, infrastructure, software and so on – and establishing who will have ownership of these.
- Organizing the transfer of software licences from one party to another.

- The physical movement of some assets – servers, for example – from one place to another.
- Migrating people's contracts, and thus terms and conditions of employment, from one employer to another.
- Dealing with severance terms for people who do not wish to move.
- Recruiting new staff to replace people who have left.
- Renegotiating any agreements with subcontractors and suppliers.

Now this is a long list and by no means exhaustive. Because of the complexity of the issues, and the fact that there are so many interdependencies involved, we are talking here about a 'programme' of work, with each element becoming an individual project. Programme management is discussed briefly in Chapter 4 but, in principle, it involves the coordination of a number of projects that, together, contribute towards some shared goal or objective.

Apart from the purely technical project management skills required for this type of IS project/programme – for example, very highly developed planning skills – outsourcing/in-sourcing affects the people involved very directly. So someone asked to manage a programme of this type must have very highly developed interpersonal skills and also the sensitivity and ability to manage the disparate groups of stakeholders involved: the organization's managers; the staff, including IT staff, affected and perhaps their trades unions too; suppliers and subcontractors; and a variety of other specialists, for example HR and lawyers.

1.9 Disaster recovery projects

We have highlighted this last type of project because, although timescale is an essential element of any project, it is especially so when there has been a large-scale failure and the organization needs to get its systems back up and running as soon as possible.

The list of things that might trigger a disaster recovery (DR) project is depressingly long, ranging from traditional causes such as fire, freezing and flood to the ever-present danger of terrorist attack. There may also be other malicious causes such as hacker attack or sabotage by a disgruntled employee and other 'accidents' such as widespread power failures. The objective of a DR project is to get the organization back on its feet as soon as possible and, as far as is practical, to ensure the continuity of the business.

The best way to manage a DR project is, of course, not to have it at all; in other words, to put in place adequate defensive measures to prevent the occurrence of the various threats. However, even the best defensive measures are not proof against everything and there can always be something that the best-prepared organization cannot have anticipated.

If a disaster *does* occur, then pre-planning of the recovery process is by far the best way of ensuring successful recovery; making things up 'on the fly' is almost certain to lead to further problems and maybe an even bigger disaster.

The following should be in place in advance and ready to be activated if needed:

- A well-thought-out disaster recovery plan, covering all the likely scenarios but also with contingency for the totally unexpected.
- Arrangements with suppliers of DR services and resources (nowadays often called 'business continuity' services), such as alternative workplaces and data centres.
- Arrangements with other parts of your organization to provide resources, such as office space, in the event of an emergency.
- Up-to-date lists of key personnel, with essential contact details.
- Equipment stored and accessible if needed, including, for example, laptops and printers.

For the manager of a DR project, the main challenges are keeping a cool head and trying to instil calm and a sense of purpose when, in fact, he or she may be having a very difficult time indeed. Panic tends to be infectious, but so does calm and reassurance and the difficulty is simultaneously to assure people that all is under control and being managed whilst at the same time instilling a sense of urgency to get things sorted out.

An excellent way of ensuring the success of DR processes is to carry out exercises to test them so that, if the real thing is needed, everyone is familiar with the 'drill'. It is a very bad time to find out that the backup server is not powerful enough and cannot be integrated into the network when a real emergency situation has developed.

1.10 Smaller IS projects

This book presents some well-established principles and techniques for managing most forms of IS project but it is sometimes asked 'is all of this really necessary for a small project?' For instance, if a developer is working on his or her own for a few days to make a small enhancement to a system, is it really necessary to go through all the rigours of estimating, planning, monitoring and reporting? In these circumstances, does all this 'project management' effort not become disproportionate to the scale and scope of the task being undertaken?

In answer to this, we would say that project management is – or at any rate should be – essentially pragmatic in nature. The approach selected should be that which will deliver the best value for money in terms of getting the job done and ensuring adequate control. So clearly, with a small project, it is both practical and sensible to adjust the project management approach to the size of the project.

One thing we feel should never be skipped is the creation of a project initiation document (PID), as described in section 7.4.3. The PID establishes, for the customer and the supplier of the project, exactly what it is about and is designed to achieve and, without a PID in place, the chances of argument and recrimination are considerably increased. The OSCAR format for a PID described in section 7.4.3 need not take up more than one or two A4 pages, or more than an hour or so to create, and yet this simple format clearly defines for all parties why the project is being undertaken. Part of the PID covers the 'constraints' on the project, including its proposed timescale, and in arriving at that the supplier will perforce have to make some estimate for the work. But this can be a very simple estimate indeed, just their 'best guess' based on their understanding of the scope/requirement – again, part of the PID – and on their knowledge of the technology to be employed.

A basic project plan is useful for supplier and customer to understand when the various tasks will be undertaken but, for a small project, this need only be a very simple bar-chart (see Chapters 8 and 10) showing the main tasks – or task if the project is very tiny.

As to all the monitoring and control, again these should be abbreviated for a small project. If, say, a project is to last three weeks, then a short email from the supplier to the customer at the end of each week should suffice to report progress and raise any issues or concerns.

So, to summarize, what we suggest about smaller projects is this: you do need some method of baselining the project and this is provided by the PID. Other typical project management deliverables, such as plans and reports, should be abbreviated so that they are proportional to the size of the project being undertaken.

1.11 Summary

There are various types of IS project that may be encountered and, whilst the general principles of managing any project are essentially the same, there are some differences in the dynamics of each type that the project manager needs to keep in mind. In addition, different types of project call for the deployment of different mixes of the project manager's skill-set – strong planning and supplier management in some cases, for example, or more developed interpersonal skills in others.

Questions

1. What are the main differences between a conventional software development project and one where a packaged solution is to be provided?
2. What are the principal difficulties facing the manager of a consultancy or business analysis assignment? How could these be overcome?
3. Why are general – in other words, not specifically IT – project management principles so relevant in the case of an IT infrastructure project?

Questions continued

4 Why is supplier and subcontractor management so important in the case of an IT infrastructure project?

5 State three reasons why an organization might decide to outsource its IT provision. Discuss the validity of these reasons.

6 What is the principal object of a disaster recovery project?

7 What adjustments to standard project management practices might be appropriate for a smaller IS project?

Case study

Introduction

The case study running through this book illustrates how the ideas and principles set out in the text could be applied in practice to IS projects. The case study itself (and the organizations referred to within it) are fictional but it does illustrate the issues and problems that the authors have encountered or witnessed in many real-life projects. We have chosen a software development project, as this enables us to illustrate most of the planning and management techniques discussed in the book but, as you will see, most of the issues are equally applicable to other types of IS project.

Background

France Vacances is a UK-based travel agency that specializes in the rental of high-quality self-catering accommodation in France. For the summer months, it offers a wide selection of *gîtes* (holiday cottages) and, for the winter, apartments and chalets in various ski resorts.

The company was founded by two friends who still own it, David Martin and Jean-Pierre Massenet, and has been in business since 1993. It has grown rapidly to achieve a turnover of some €6.75 million per annum and employs 85 staff at two offices, one in the Greater London area and one in France.

France Vacances currently uses two main sales channels:

- Direct selling to customers through mailshots of its brochures and customer support centres (70 per cent of sales).
- Sales via high street travel agents (30 per cent of sales).

However, the company is aware from press coverage and from surveys among its own customers that there is a growing public demand to be able to book holidays via the internet. This is particularly true as its customers are precisely the sort of people who are 'net aware'. France Vacances does

Case study continued

have a website but this is really just its latest brochure in electronic format and it does not have links to up-to-date availability data or the facilities for customers to make secure bookings online.

Consequently, France Vacances has decided to implement a new internet-based booking system. This will be linked to its existing computerized booking system, which contains data on the availability of properties, and to its customer database as well as having secure links over which credit card data can be received. In addition, the company wants its management information system (MIS) enhanced so that it can trawl its databases and send targeted information to customers on properties that are likely to be of interest to them.

France Vacances organization

The current organizational structure of the company is shown in Figure 1.1. In essence, the two founders have divided the business between them. David Martin (who has a sales background) looks after the sales and operations side, and Jean-Pierre Massenet, an accountant, takes care of finance and administration.

The small IT department within the administration function consists of the IT manager, Peter Clay, three analyst/programmers and a computer operator/trainee programmer.

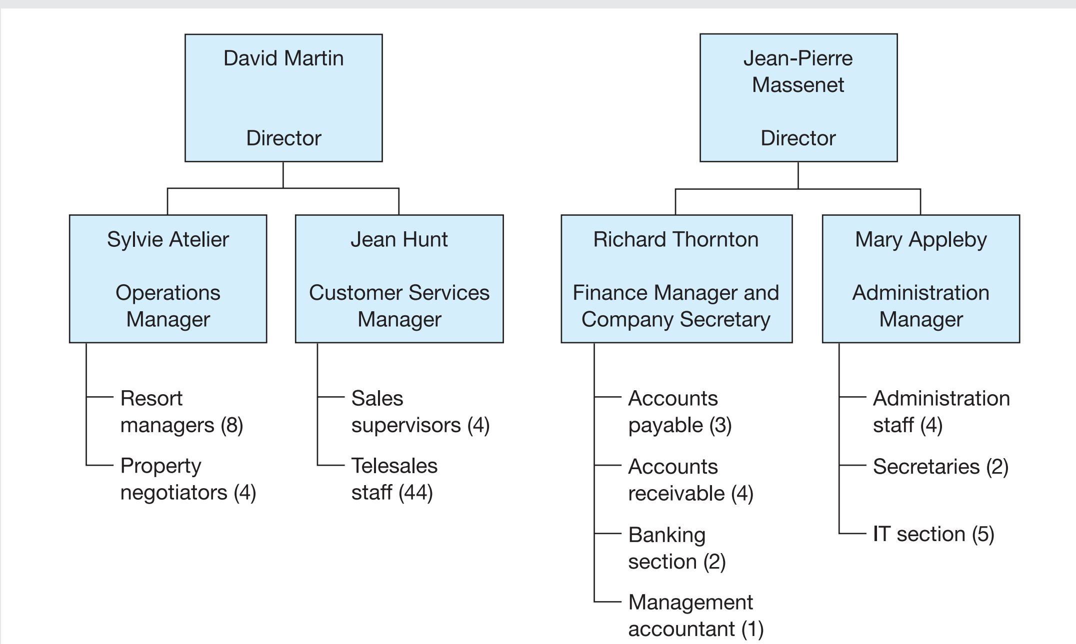

Figure 1.1 Organization of France Vacances

Case study continued

The project

Because of the small size of its IT department, and since the department lacks skills in the design of e-commerce applications, France Vacances has decided to entrust the development of its internet service to a consultancy company, E-Con. This firm has tendered for the following services:

- Analysis of the requirements.
- Production of a detailed requirements specification.
- Design, development and implementation of the internet systems, including a new website and secure communications links.
- Training France Vacances staff in the use of the new systems.
- Specification of the interfaces required from France Vacances's existing customer database and booking system (the development of the links at the France Vacances end to be done by its own IT department).
- Specification of the additional hardware required to support the new system (to be obtained from France Vacances's usual suppliers, the procurement to be managed by the IT department).
- 'Skills transfer' to France Vacances's IT department, so that ongoing maintenance and development of the new system can be handled in-house. The development of the MIS aspects of the new system will be dealt with by France Vacances's IT department.

The date now is 1 April and France Vacances wants to have the new system up and running for the start of the winter season's bookings at the end of June.

Further reading

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